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The impact of using an IoT-based automatic solar panel cleaner on building energy performance

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Abstract

Solar energy is a great provider of renewable environmentally friendly energy. Thanks to solar technologies, it converts sunlight into electrical energy using multiple devices such as photovoltaic cells. The generated electrical energy can be used for multiple purposes such as commercial, industrial, and domestic uses. Generating electrical energy without releasing toxic gases is considered one of the biggest advantages of using solar energy. While solar panels are a reliable source of clean electrical energy, their power output can be decreased due to the lack of maintenance and especially cleaning. The efficiency of solar panels decreases by more than 40% due to the lack of cleaning. That is why routine cleaning is crucial to ensure the optimum performance of them. Providing an automatic solar panel cleaning system will ensure that routine maintenance is done and the electricity output is optimum. Integrating the use of photovoltaic cells in buildings helps improve the energy performance and energy efficiency of buildings. This research paper aims to determine the impact of integrating an internet of things (IoT)-based automatic solar panel cleaner and measure the energy efficiency of using it. The IoT-based automatic solar panel cleaner contributed to raising the solar panel power wattage by 89.33% in a hot desert climatic region.

Keywords: solar energy; office buildings IoT; solar panels

1 Introduction

In light of the world's energy issues, the depletion of fossil resources and our extreme reliance on power have made it imperative for governments and individuals to find renewable energy sources. The construction sector is responsible for over 40% of total energy consumption, which is a huge value specifically in a country like Egypt that depends on fossil fuel as the main source of energy [1]. Renewable energy sources are necessary to facilitate the switch to clean, sustainable energy sources. Among the principal renewable energy sources are hydropower, wind, sun, geothermal, biomass, and soil. Solar energy is favored above other renewable energy sources because of its abundance, advantages for the environment, versatility, scalability, and cost-effectiveness. Solar technology is becoming more and more popular as a clean and sustainable energy source due to its ongoing developments, being pollution-free, and falling costs [2]. Due to the technology's rapid development and innovation and the declining cost of its component elements, solar energy has gained acceptance across all economic sectors. Solar panels systems are becoming more and more in demand every day since they are used in industrial and agricultural activities, as well as in lighting, heating, cooling, and household appliances. Solar panel systems have more diversified applications because they provide electrical energy, which is both more in demand and easily convertible into other forms of energy. In the upcoming years, it is anticipated that there will be a major increase in solar installations worldwide. According to BloombergNEF, a leading provider of forward-thinking primary research and analysis on the trends driving the transition to a lower-carbon economy, there will be a 56% rise in solar installations worldwide in 2023 over the previous year. Approximately 92 GW of solar power was installed worldwide in the second quarter of 2023, with China accounting for 55% of that total. According to Taiwanese market research firm TrendForce, there is a possibility for the installation of 350.6 GW of solar panel systems worldwide in 2023, which would be a 53.4% increase from 2022. The US solar sector anticipates record-breaking 32 GW of new capacity added in 2023, up 52% from 2022. Europe's solar installation levels are exceeding expectations with a plan to add 58 GW and are on track to grow by 30% this year. Moreover, the Middle East and North Africa (MENA) region is embracing solar power and experiencing growth in solar installations. The growth is being driven by favorable government policies, falling costs, and an abundance of sunlight (Fig. 1). The MENA area is anticipated to witness notable expansion. The MENA area is well known for having substantial oil and gas reserves and for playing a pivotal role in the world energy markets. On the other hand, the area leads in terms of energy intensity and utilization. Because of the direct normal irradiance (DNI) of more than 5 kWh/m^2 per day, solar electricity generation is viable in over 85% of the MENA. In the MENA region, solar electricity is well accepted by the community due to its perceived environmental friendliness. The MENA area, which includes Iran, the United Arab Emirates, Egypt, Jordan, Morocco, and Egypt, has set itself lofty goals for adding renewable energy to its energy mix. An EU-MENA energy grid connection is

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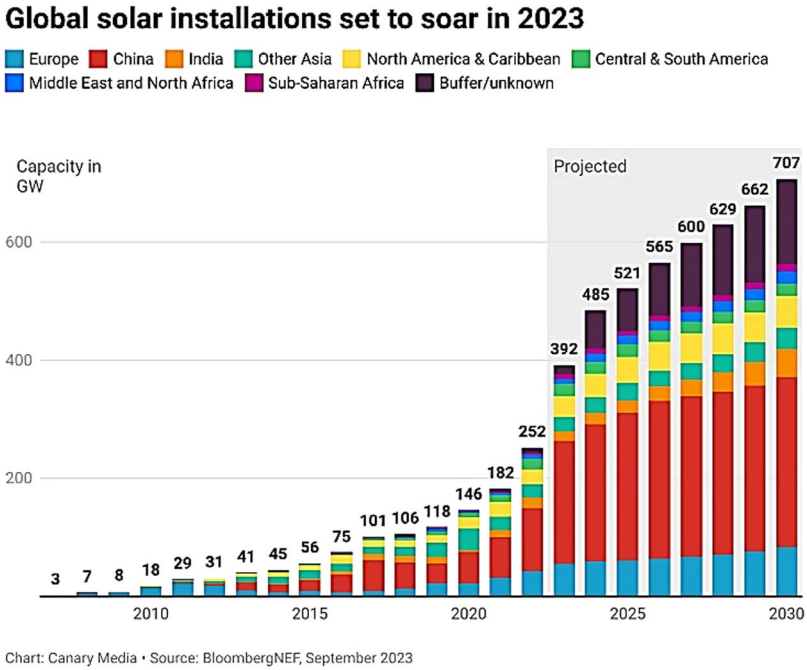


Figure 1. Projections for solar panel installations worldwide by 2023.

becoming more and more popular because of the region's solar potential, advances in concentrating solar power technology, and the need to cut carbon emissions. In the MENA region, the switch from fossil fuels to solar power is viewed as a logical step, with solar technology being essential to desalination and sustainable energy. A number of MENA nations, including Saudi Arabia, Abu Dhabi, Dubai, Egypt, and Morocco, have achieved notable progress in the development of renewable energy. Notable developments include record-breaking solar prices and the generation of jobs through the growth of local industries.

The creation of effective methods for exploiting renewable energy is crucial for the Mediterranean region's sustainable growth, given its abundant supply of renewable energy resources. And as stated in Atlas Solar Radiation of Egypt, the annual direct solar radiation of Egypt is between 5.4 and 7.1 kWh/m^2 and the sufficiency of solar potential is nearly 74 billion MWh/year which is considered a promising investment economically and environmentally as it mitigates carbon dioxide emissions and substitutes the use of fossil fuel [3]. Photovoltaic cells are not widely used in the MENA region compared to the high solar radiation availability throughout the year. Table 1 shows the annual DNI, annual sunshine duration in some MENA capitals, and annual global horizontal irradiance (GHI) in comparison to some European capitals where photovoltaic cells are more widely used and the sunshine hours and solar irradiance are almost half of that at the MENA region.

The Egyptian government, in keeping with its 2030 development strategy, has urged big investors to help construct power plants and small investors and local communities to incorporate solar energy into their residential buildings. However, a significant issue with solar panel systems in Egypt and other regional settings is that their performance deteriorates when exposed to various environmental factors such as high temperature, snow, sea salt, wind, dust, dirt, bird's dropping pollen, clouds, and shade. This has prompted researchers to investigate mitigating strategies. The performance and efficiency of photovoltaic solar panels can be adversely affected by dust collection. One of the main issues of using a solar panels system is the airborne dust that accumulates on the panels. Dust prevents collecting solar radiation and obscures solar flux, which reduces the efficiency of the cells. To ensure the efficient operation of panels, dust should be removed [4]. Dust acts as a barrier when it collects on the panel surface, lowering transparency and obstructing sunlight from reaching the solar cells. The outcome is a drop in the solar panels' output power and efficiency, which lowers performance and can shorten the panels' lifespan. Globally, there are differences in the amount of dust, but arid regions like the MENA have some of the greatest amounts. It is interesting to note that these areas get a lot of solar radiation as well. In desert regions, dust buildup on photovoltaic panels can be especially severe; research indicates that this can result in a 40% reduction in solar efficiency. Dust storms are common in Egypt, especially in specific seasons [5]. The efficiency of a solar panel system installed in Riyadh, KSA, a hot arid region, resulted in a reduction of 32% in output energy varying between clean cells and unclean cells [6]. Due to the decrease in solar panel cells' efficiency over time in Egypt, the MENA region, and similar locations, current maintenance and cleaning approaches are labor intensive and inefficient. This research paper suggests an internet of things (IoT)-based, microcontroller-operated automated solar panel cleaning and power monitoring system. This is because microcontrollers are highly suitable in many application fields, and IoT is a state-of-the-art technology for any monitoring system. In this paper, a design of a flexible cleaning tool that runs the whole length of the solar panel and IoT mobile applications can be utilized to trigger cleaning instructions as necessary. The results demonstrated that external resistance could reduce the performance of the solar panel by up to 40%. The rest of this paper is arranged as follows: Section 2 summarizes the cleaning methods and techniques in the literature review. Section 3 provides a general classification

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Table 1. Geographical and meteorological data of some MENA capitals and European Capitals [1]

Location		GHI (kWh/m^2)	DNI (kWh/m^2)	Sunshine (hours)	PM2.5 (μg/m^3)
MENA locations					
Emirates, Abu Dhabi	24.4539° N, 54.3773° E	2171	1888	3609	41
Egypt, Cairo	30.0444° N, 31.2357° E	2010	2039	3348	87
Jordan, Amman	31.9544° N, 35.9106° E	2122	2402	3290	33
Iran, Tehran	35.7219° N, 51.3347° E	1928	1940	3010	39
Kuwait, Kuwait City	29.3759° N, 47.9774° E	2032	1830	3000	61
Morocco, Rabat	34.0084° N, 6.8539° W	1843	1777	2916	33
European locations					
Czech, Prague	50.0755° N, 14.4378° E	1117	995	1668	16
Germany, Berlin	52.5200° N, 13.4050° E	1062	964	1625	12
UK, London	51.5072° N, 0.1276° W	1023	866	1574	10

of the cleaning techniques, while Section 4 provides details of the methods and materials, which include the proposed IoT-based automatic solar panel cleaner system and a case study. Section 5 delves deeply into the results and analysis; finally, the paper concludes with Section 6 highlighting recommendations for future work.

2 Literature review

Several parameters, such as the density, content, and particle size distribution of the deposited dust, affect how well solar panels operate when it comes to dust. Different types of dust accumulated from diverse sources can have variable effects on solar panel module efficiency reduction. Consequently, the degree to which dust has an impact on solar panel performance can vary depending on its unique properties in a given area. Since natural dust buildup is a site-dependent phenomenon rather than a global one, it has been demonstrated that MENA countries are the most affected, with regions such as parts of Asia following closely behind (Fig. 2). Some dust can be removed from the panels by wind and rain, but in some situations, the dust may become firmly embedded in the surface, necessitating more intensive cleaning techniques. Solar panels can be kept operating efficiently and provide optimal energy generation with proper cleaning. It is advised that solar panels be cleaned regularly to reduce the damaging effects of dust accumulation. As a result, a great deal of research was done on solar energy and improving the efficiency of solar modules. A vast amount of material has been written about solar panel soiling and cleaning to comprehend how soiling affects solar panel modules, forecast efficiency reductions brought on by soiling, and find solutions to the soiling issue.

This research focuses on the MENA, namely Egypt. Many studies have examined this influence and the resulting drop in solar panel efficiency in MENA's most afflicted countries. A town in Riyadh, Saudi Arabia, that uses a solar panel system was identified as a suitable site for studying this phenomenon. Mani and Pillai [7] and Salim *et al.* [8] investigated the dust collection impact for 8 months and found a 32% loss in solar panel performance. Another location is in Kuwait City, where Wakim [9] discovered a 17% drop in electricity after only 6 days due to sand collection on panels. Sayigh also investigated the effect of dust deposition on tilted glass plates with tilt degrees ranging from 0° to 60° and found that after 38 days of exposure, plate transmittance decreased by 17% to 64%. In addition, Sayigh *et al.* [10] found that after 3 days of dust deposition, the usable energy gain of a horizontal collector decreased by 30%. Kazem *et al.* [11] evaluated the influence of dust on the performance of (polycrystalline) solar modules in Oman's desert climate. Their findings reveal that the decline in panel performance is dependent on the pollutant type and deposition amount. The biggest fall in solar panel voltage was caused by ash pollution (25%). In Bangladesh, Rahman *et al.* [12] discovered that dust deposition on the surface of a photovoltaic solar panel system affected efficiency by up to 35% during a 1-month period. Furthermore, following 10 days of exposure at Roorkee, India, Shaharin *et al.* [13] showed an average of 8% drop in glass plate (tilted at 45°) transmittance. Ali *et al.* [14] conducted an experiment in Taxila, Pakistan, using two different types of photovoltaic panels (monocrystalline silicon and polycrystalline silicon) and two modules of each type, with one module from each pair exposed to the natural atmosphere for 3 months during the winter to investigate the effect of dust deposited on the surface of photovoltaic modules. A systematic set of measurements was performed matching to the various dust densities. The difference between the output characteristics of clean and dirty modules revealed the percentage loss at various dust densities. After 3 months, the dust density on the module's surface was 0.99 mg/cm^2. The average output power of monocrystalline and polycrystalline modules decreased by around 20% and 16%, respectively, as compared to clean modules of the same kind. The efficiency drop for a monocrystalline and a polycrystalline module was 3.55% and 3.01%, respectively. Tanesab *et al.* [15] conducted an investigation on photovoltaic modules at Murdoch University in Perth, Australia, for 18 years without cleaning procedures, measuring the contribution of dust to the long-term performance degradation of various photovoltaic solar panel modules. The solar panel module's power output deterioration varied from 19% to 33%. The deterioration is primarily caused by non-dust-related processes, accounting for around 71–84% of the total, yet dust still contributes significantly (16–29%). Over a 2-month period (August–September 2009), an experimental investigation was conducted to measure the influence of urban air pollution on the energy performance of solar panels [16]. Kaldellis *et al.* [17] conducted their experiment in Athens, Greece. According to the findings, solar panels produced up to roughly 6.5% less energy than a clean panel.

Hee *et al.* [17], from Singapore's Department of Electrical and Computer Engineering, discovered that bare glass transmission was decreased despite significant rains over several months. After 33 days, the transmission of simple glass slides

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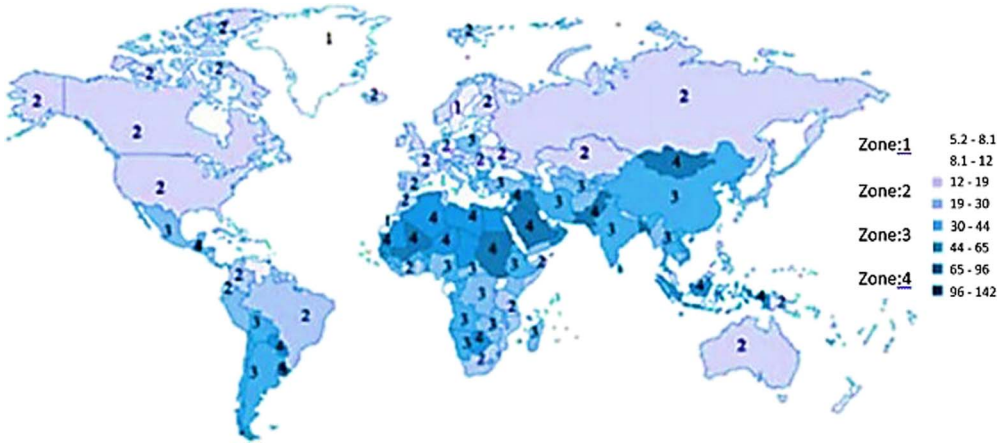


Figure 2. Dust intensity around the world [7].

decreased from 90.7% to 87.6%. The performance of TiO2 coating on solar panels with various film thicknesses (60 and 40 nm) was examined [18]. The hydrolysis technique was used to deposit TiO2 films, which is a low-cost and commercially feasible procedure. It was discovered that TiO2 coatings indeed aid to lessen the influence of dust on transmission, with a thicker 60-nm coating being more effective than a thinner 40-nm coating. According to Ref. [18], outdoor tests were done at the national energy research center in Soba, Khartoum State, Sudan, using two silicon monocrystalline panels (washed before each measurement and unwashed) exposed to natural sunlight for 3 months. The results demonstrate that dust deposition affects the efficiency of solar panels by 4.3% to 16.8%.

Cleaning solar panels is crucial for improving their output power and efficiency [19]. Researchers have proposed and researched various cleaning methods for photovoltaic modules [20]. Manually washing the plate with water and a soft bristle brush is one method, but it is time-consuming and requires specific tools. Previous research compared the performance of manually cleaned solar panels to those cleaned by a robotic system and discovered that the robotic system was more efficient in terms of energy harvesting [21]. In addition to cleaning, monitoring the output of solar modules is an important part of solar module maintenance. Several studies have demonstrated that real-time monitoring of solar panel performance can aid in the identification and resolution of issues such as shadowing, pollution, and other performance-related concerns [22]. Kumar *et al.* [22] presented a wireless sensor network-based system for tracking solar panel performance. This is useful in detecting defects and increasing overall system efficiency. An artificial intelligence-based solar panel cleaning system was presented in a research by Zhou *et al.* [22]. It makes use of machine learning algorithms to determine the degree of soiling on the panels and adjust the cleaning procedure accordingly. According to the study, the suggested solution outperformed conventional cleaning techniques in terms of both effectiveness and affordability [23].

An IoT-based solar panel cleaning and monitoring system that combines sensors, actuators, and machine learning algorithms to identify and resolve problems instantly was presented in a different study by Tewari *et al.* [23] According to the study, the suggested solution effectively increased energy production while lowering maintenance expenses. A solar panel cleaning and monitoring system that makes use of drone technology to clean and inspect the panels was suggested in a study by Bhat *et al.* [24]. According to the study, the suggested approach effectively decreased cleaning times and increased inspection accuracy. Some research has concentrated on creating self-cleaning solar panels that can lower maintenance needs in addition to cleaning and monitoring. A hydrophobic coating is used in a study by Kippelen *et al.* [25] to create a self-cleaning solar panel that keeps dust and water away. According to the study, compared to conventional panels, the suggested panel was more efficient and had a longer lifespan.

Overall, prior studies have demonstrated that keeping solar panels clean and well maintained is crucial to maximizing their lifespan and performance. An effective, economical, and automated solution for solar panel dust detection and cleaning techniques is offered by the proposed IoT-based automated solar panel cleaning, which attempts to solve these problems.

3 Solar panel cleaning techniques

According to Aljaghoub *et al.* [26], solar panel cleaning is an important issue concerning the efficiency and longevity of the cells due to being exposed to climatic and weather conditions. There are multiple cleaning techniques that are natural cleaning, manual cleaning, automated cleaning, and electrostatic cleaning. Natural cleaning depends on rain and wind to clean the panels, which relies on the frequency of both of them. Manual cleaning techniques using water, detergent, and brushes are a feasible cleaning technique, yet they require a large number of labor. The problem with this technique is that labor must be well trained in order not to cause scratches on the panel, and the bigger the size of the panel, the more challenging the cleaning process is. In order to reduce labor costs, automated cleaning techniques were devised. The automated systems are reliable but they raise the operational and maintenance costs. Electrostatic cleaning uses electrostatic force to clean panels with a touchless method to avoid scratches; however, this system is very expensive. Table 2 lists the parameters for comparing solar panel cleaning methods: manual cleaning, automatic cleaning, and electrostatic cleaning. The parameters include multiple variables that determine their advantages and disadvantages.

Both automatic cleaning techniques and electrostatic techniques have better parameters, yet the high cost of electrostatic cleaning techniques is a great barrier to using them in

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Table 2. Parameters comparing solar panel cleaning methods: manual cleaning, automatic cleaning, and electrostatic cleaning [28]

Parameter	Manual cleaning	Automatic cleaning	Electrostatic cleaning
Cost	Low	Moderate	High
Efficiency	Low	High	High
Cleaning time	Long	Moderate	Moderate
Operation effort	High	Low	Low
Scratches opportunity	High	No	No
Output power	Moderate	High	High
Human dependent	Yes	No	No

developing countries like Egypt. Our proposed system not only took advantage of automated cleaning techniques with its no labor advantage and autonomous operation but also added the IoT-based system that utilizes advanced technology to enable devices to communicate by exchanging data over the internet to control and monitor it. Our proposed system automates the cleaning process based on multiple instantly changing parameters such as weather conditions. It also follows predefined cleaning schedules. It detects the level of dust and dirt on panels and starts cleaning automatically without any need for labor or manual intervention, which optimizes the automatic cleaning standard systems.

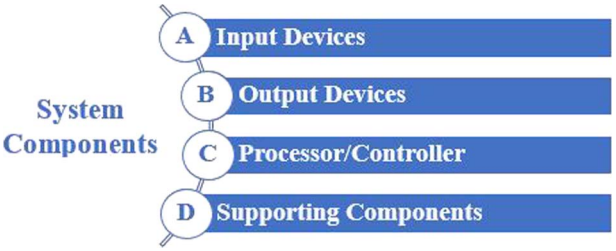


Figure 3. Major components of the proposed IoT-controlled solar panel cleaner system.

4 Materials and methods

4.1 Proposed IoT-controlled system for cleaning solar panels

Solar panels are an effective and environmentally beneficial way to produce power from sunshine. However, the collection of dust and grime on the surface of solar panels can dramatically limit energy generation efficiency. To overcome this issue, an IoT-controlled solar panel cleaning system was created. This IoT-controlled solar panel cleaner system is an effective way to maintain the performance of solar panels. The system uses IoT technology for remote monitoring, sophisticated analysis, and automatic cleaning, resulting in higher energy generation and overall efficiency. Regular cleaning of solar panels is required to guarantee peak performance and energy output.

4.1.1 System architecture

An IoT-controlled solar panel cleaning system is recommended for removing dust from the solar panel's surface. According to the size of the flat plate panel, the system comprises a brush driven by a DC motor through a threaded rod system. The microcontroller creates a signal depending on the data from the dashboard button, causing the brushes to rotate. Aside from rotational motion, stepper motors receive a signal that is converted into linear motion via a rod. The brush then glides in a linear motion up and down the surface of the solar panel. The key components of the suggested cleaning technique and their roles are divided into four categories (Fig. 3).

A) Input Devices

– Voltage Sensor: detects and measures electrical potential difference (voltage) in a circuit. It provides data about the voltage level in the system.
– Current Sensor: Similar to the voltage sensor, the current sensor measures the flow of electric current in a circuit, providing information about current levels.

B) Output Devices

– DC Motor: They produce mechanical motion (rotary motion) in response to electrical signals. They convert electrical energy into mechanical energy.
– Stepper Motor: They produce precise motion (linear motion) in discrete steps, making them suitable for applications requiring accurate control over position and speed.

C) Processor/Controller

– ESP32 Microcontroller: It serves as the brain of the system, handling tasks such as data processing, decision-making, and communication with other devices. It integrates Wi-Fi and Bluetooth capabilities, making it suitable for IoT applications.

D) Supporting Components

– Voltage Regulators: They are part of the power management system, ensuring stable voltage supply to other components. They regulate voltage and are more closely associated with the power supply subsystem.
– Relay Module: They are used to control high-power devices with low-power control signals, often from a microcontroller like the ESP32. They facilitate communication between the microcontroller and high-power output devices.
– Motor Drivers: An electronic device that controls the motion and direction of a motor by regulating the current supplied to it. It acts as a bridge between the microcontroller and the motor, allowing the motor to operate efficiently and perform specific tasks.

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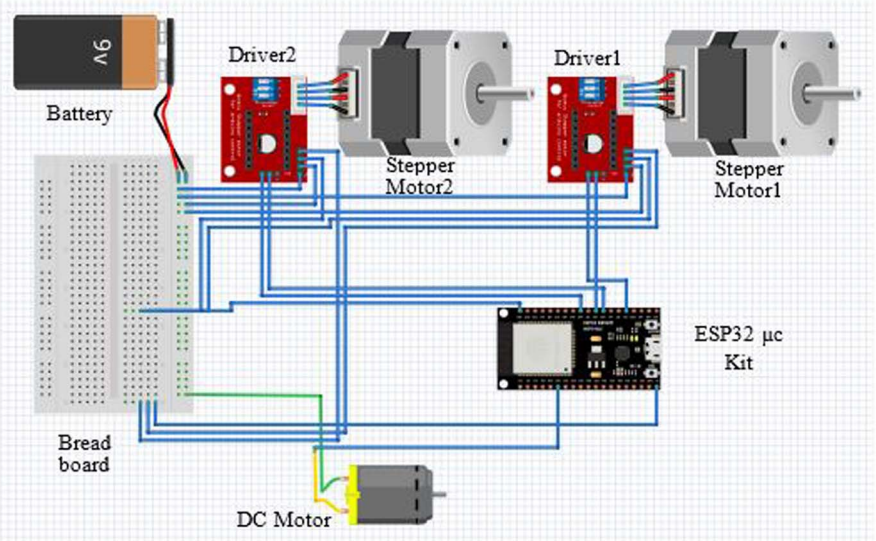


Figure 4. Schematic diagram of IoT-controlled solar panel cleaner system using Fritzing simulator.

A schematic diagram of the proposed IoT-based automated solar panel cleaning and monitoring system in the Fritzing simulation environment is shown in Fig. 4.

4.1.2 System operation

The operation of an IoT-controlled solar panel cleaner system involves the integration of IoT technology to monitor, control, and automate the cleaning process of solar panels. Here is a breakdown of the operation of the proposed IoT-controlled solar panel cleaner system:

1) Dust Monitoring: Sensors provide data on the system's voltage and current levels, which may be interpreted as showing the deposition of dust and filth on solar panel surfaces. These sensors give real-time data on dust levels, which is important for determining when to clean.
2) Wiping System With Motor Control: The IoT-controlled solar panel cleaner uses a wiping system with motor control to clean the solar panel surface. The wiping mechanism removes dust and grime from the panels.
3) Remote Monitoring and Control: The system is built for remote monitoring and control. Users can connect to the system via an application or web interface from anywhere (Fig. 5a). This enables real-time monitoring of the cleaning process and intelligent decision-making based on the data acquired by IoT sensors.
4) Cleaning Process: When the voltage and current measurements fall below the threshold value (which is 20% less than the average value), the dashboard app sends a cleaning launch command to the microcontroller. The controller will send control signals to the system to initiate cleaning. The DC motor receives a signal to rotate the brush, whereas the stepper motors get signals to scan the solar panel "brush motion linearly up and down." The brush begins its travel from the home position and moves forward. After a given number of steps, the brush reaches the panel's ends, known as "the end position." After this moment, the motors (DC and stepper motors) come to a halt, while the stepper motors reverse their motion and begin to move in the other direction. The brush movement continues until it returns home (Fig. 5b–d).
5) Efficiency Improvement: By regularly cleaning the solar panels, the system helps maintain their optimal performance and efficiency. Accumulated dust and dirt can significantly reduce the energy generation capacity of solar panels, so cleaning them improves their effectiveness and increases energy output. Section 4 of the outcome analysis discusses this improvement in further depth.
6) Cost-effectiveness and Scalability: The proposed IoT-controlled solar panel cleaner system is considered cost-effective and scalable compared to traditional labor-based cleaning methods. It reduces the need for manual labor and provides an automated and efficient cleaning solution.

The flow chart of the proposed IoT-based automated solar panel cleaning system is depicted in Fig. 6. According to the flow chart, the software initializes the variables upon system startup. It initiates the microprocessor and ensures that the stepper motor and brush are in the starting position (home). Then it connects to the internet. Then it examines the microcontroller's input ports for current and voltage sensor data. Following that, the data are sent to the mobile dashboard via the WiFi module. The data value is examined. If the reading is less than the threshold value (20%), a command signal is sent to begin the cleaning procedure; otherwise, the operation ends. The cleaning operation begins at the top of the module, with the DC motor turning the brush in an angular motion and the stepper motors moving the brush linearly forward. The cleaning procedure continues until the brush reaches the end of the panel (the bottom of the module), at which point the motors stop rotating, the stepper motors reverse direction, and the DC motor restarts. The cleaning procedure continues in the opposite way until it reaches home (the top of the module) again. The motors and the whole system stop the cleaning operation once it is done.

4.1.3 Benefits of the proposed system

The IoT-controlled solar panel cleaner system has various benefits over traditional labor-based cleaning systems, which

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Figure 5. The proposed IoT-controlled solar panel cleaner system, (a) the dashboard app, (b) the brush at rest (home position), (c) the brush at the middle of the panel during the cleaning process, and (d) the brush at the end of the forward motion and the start to the reverse motion.

are expensive, are time-consuming, and lack monitoring capabilities. Here are some of the advantages of the proposed system:

- The automatic cleaning solution reduces the need for manual effort.
- IoT technology improves solar panel efficiency by detecting dust, analyzing data, and controlling the system.
- The system removes dust and dirt from solar panels, ensuring maximum performance and energy generation.
- Regularly cleaning solar panels improves their efficiency and effectiveness, leading to greater energy output.
- IoT technology enables remote monitoring and management of the system, enabling informed cleaning decisions, and making it easier for system owners to manage cleaning.
- The IoT-controlled solar panel cleaning system is a cost-effective and scalable solution for removing soil and dust from solar panels.

4.2 Case study

In this section, a detailed case study will be presented, as well as the climatology of the case study zone, the building description, the base case simulation, and the impact of installing photovoltaic cells on the buildings and how this leads to energy savings.

4.2.1 Climatology of the case study zone

The Köppen climatic classification system assigns Egypt's hot desert climate a BWh designation. In Egypt, dust storms are frequently linked to the khamsin, a scorching wind that blows sand and dust all over the place. Khamsin storms can last for several days and usually happen between March and May. Large amounts of sand and dust from the deserts can be carried by these storms, affecting daily life and decreasing visibility. Temperatures can sometimes climb by 20°C in a matter of hours due to the khamsin wind, which can reach velocities of up to 140 km/h. It is crucial to remember that dust storm frequency and intensity can change from year to year and between Egypt's various regions. Reliable sources of local weather forecasts and updates can offer more precise and current information about Egypt's dust weather conditions.

Table 3. Simulated monthly energy consumption of Building "47"

Month	Energy consumption (kWh)
January	66 543
February	68 420
March	91 355
April	109 398
May	130 456
June	164 662
July	190 977
August	190 978
September	174 436
October	157 142
November	104 135
December	77 819

Fig. 7 depicts the impact of Khamsin storms on several regions in Egypt.

4.2.2 Case study building description

The case study was applied to an office building located in New Cairo, Egypt. The building is "47" Office Building that is located in Street 90-North. The building consists of seven floors. The building footprint is 1735 m^2. The building is located in (30° 01′ 14.45″ N, 31° 186 25′ 28.65″ E) Cairo, Egypt, as shown in Fig. 8 in the hot arid climatic zone.

4.2.3 Base case simulation

The building is simulated and modeled using DesignBuilder Software and is simulated using the location template CAIRO INTL AIRPORT (Fig. 9). The building energy consumption is simulated using Energy Plus software through DesignBuilder V7. Building data are inserted and modeled such as the activity of spaces, HVAC systems, glazing, and operation hours. The building is modeled and simulated to check the required annual energy to be met partially or totally using photovoltaic cells.

The energy simulation of Building "47" on DesignBuilder software resulted in the base case model annual energy consumption = 1 526 321 kWh, and the monthly energy consumption is as shown in Table 3.

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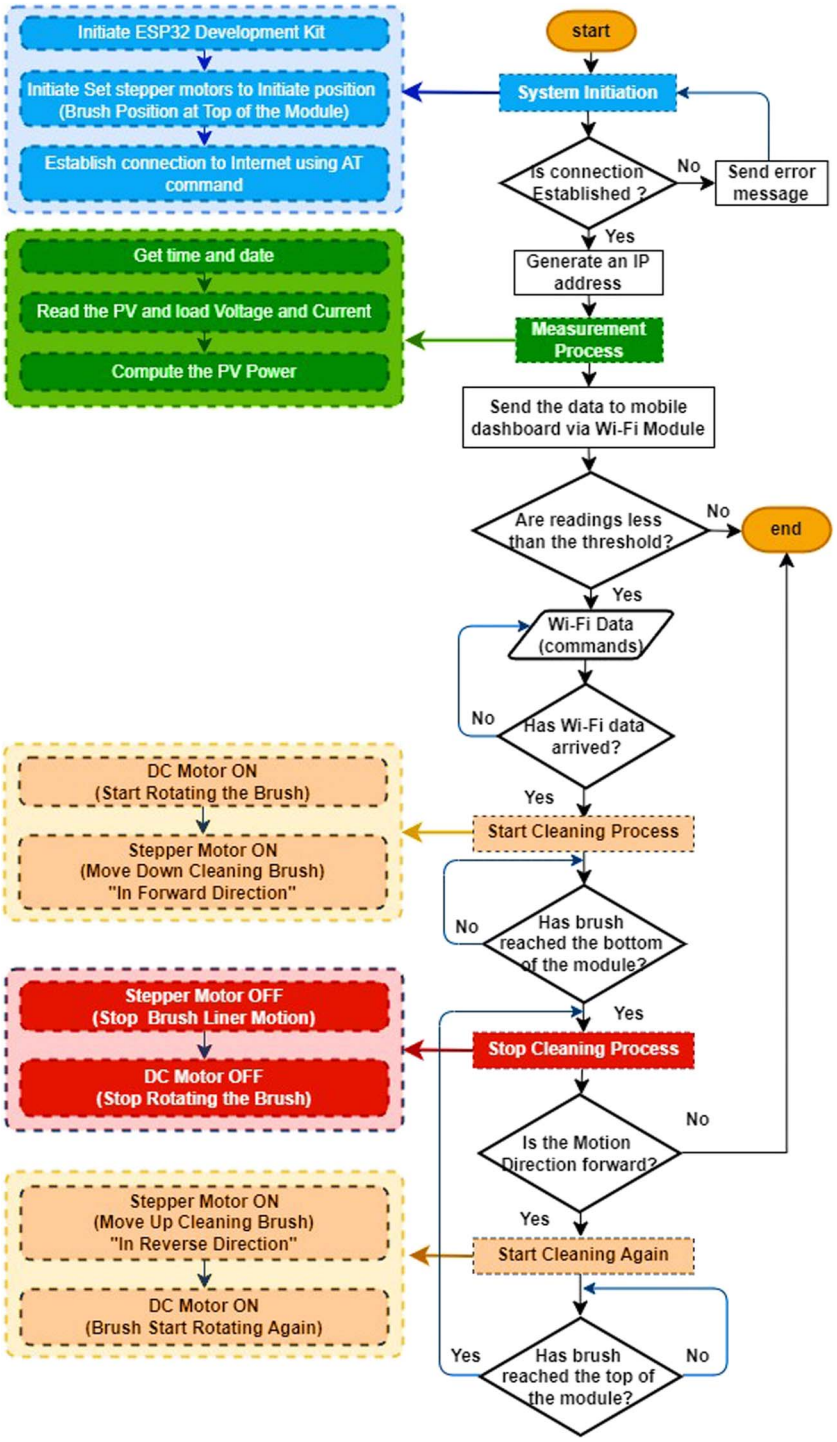


Figure 6. The flowchart of the proposed IoT-controlled solar panel cleaner system.

4.2.4 Applying photovoltaic cells to the buildings

Using photovoltaic cells on the roof of the case study building involved using photovoltaic module, a polycrystalline silicon cell with 20% efficiency. The specifications of the photovoltaic modules are shown in Table 4. The photovoltaic modules will cover the roof area, which is 1735 m^2.

4.2.5 Photovoltaic system output calculations

The expected output of the photovoltaic modules is calculated. According to Omran [27], the normal incidence beam irradiation for Cairo, Egypt's, annual average daily value is 16.94 MJ/m^2 for Cairo. According to the Global Solar Atlas, Egypt receives an average of 3050 h of sunlight per year, having the average DNI that ranges from 1970 to 3200 kWh/m^2 [28]. The average DNI will be 2585 kWh/m^2. According to Robaa [29], the sunshine duration ranges between 9 and 11 hours in Cairo Egypt. Equations (1–4) show the calculations of the output of solar system applied to the building.

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Figure 7. The impact of khamsin storms on several regions in Egypt.

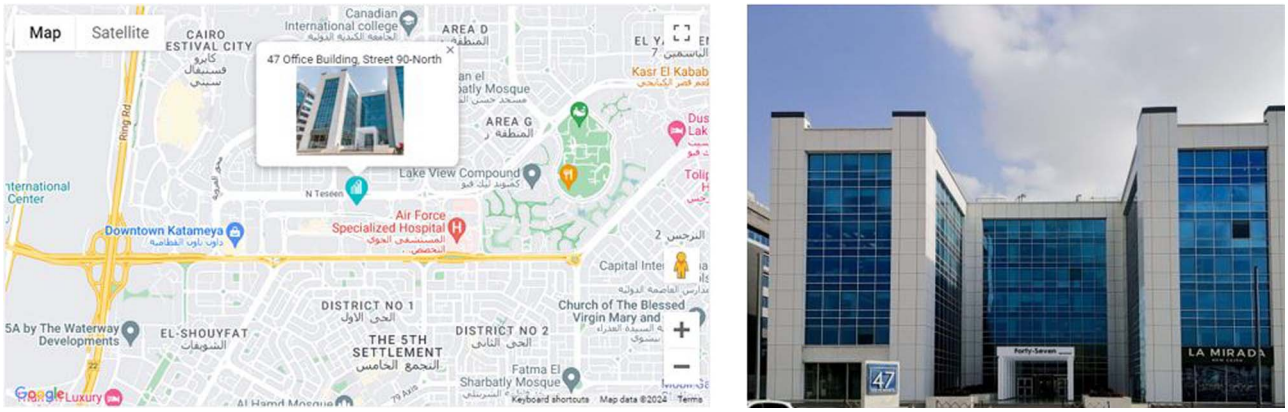


Figure 8. The case study office building in Cairo. (a) The location of Building "47" on a map. (b) Elevation view of Building "47."

$$\text{Area Covered by Panel (m}^2\text{)} = \text{Panel Length (m)} \times \text{Panel Width (m)} \quad (1)$$

$$\text{Electricity Output (kWh)} = \text{Power Rating (kW)} \times \text{Sunlight Duration (hours)} \times \text{Efficiency} \quad (2)$$

$$\text{Total Daily Output (kWh)} = \text{Electricity Output per Panel (kWh)} \times \text{Number of Panels} \quad (3)$$

$$\text{Annual Output (kWh)} = \text{Total Daily Output (kWh)} \times 365 \text{ days} \quad (4)$$

According to equation (1), the area covered by each panel is 0.1575 m^2; hence, 11 015 panels cover the building roof of 1735 m^2. The output power of the panel with a 20% power rating is 0.04 kWh when exposed to sunlight for 10 hours, and the panel's working efficiency reaches a high of 20%, as estimated by equation (2). Using equation (3), the total daily output electricity of the panels is 440.6 kWh. Equation (4) provides a simple expression for the annual output. As a result, the specified solar system will give 160 819 kWh,

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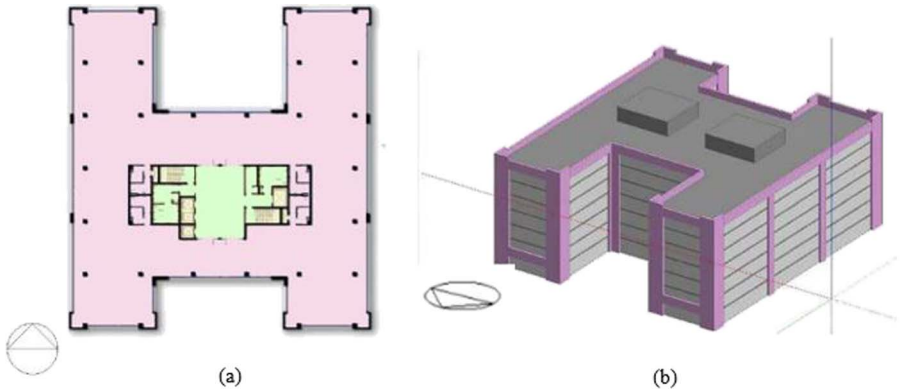


Figure 9. The building simulation. (a) Building "47" floor plan on DesignBuilder software. (b) Isometric view of the Building "47" model on DesignBuilder software.

Table 4. Specifications of the used photovoltaic cells

Specification	Value
Dimensions ($L \times W \times H$)	450 mm × 350 mm × 17 mm
Rated power output	20 W
Efficiency	20%
Cell type	Polycrystalline silicon
Maximum power voltage (V_{mp})	18 V
Maximum power current (I_{mp})	1.11 A
Open circuit voltage (V_{oc})	21.96 V
Short circuit current (I_{sc})	1.16 A
Maximum series fuse	5 A
Maximum system voltage	1000 V
Power selection	±3%

whereas the DesignBuilder simulation indicates that the building requires 1 526 321 kWh. The installed photovoltaic cells will provide the building with almost 10.5% of the required energy.

4.2.6 On-site experimental investigation

This part of the study is responsible for investigating the dust and other pollutants accumulating on the solar panels by exposing two solar panels to natural weather conditions. One solar panel will be cleaned using our IoT-based automatic solar panel cleaner, and the other solar panel will be left without cleaning. According to Mayhoub *et al.* [30], it is recommended to apply cleaning to solar devices that are exposed to weather conditions, especially dust in the case of hot desert climates like Cairo, once every 2 weeks to keep the efficiency up to 90%.

5 Analysis and results

Dust influences photovoltaic module efficiency, which in turn impacts solar panel performance, yield, and profitability. The maximum output of a photovoltaic panel coated in dust was lowered when compared to a clean solar panel. Covering the panel with dust diminishes its power and efficiency because the dust acts as a barrier between solar energy and the panel.

The thickness of the dust layer affects the performance of solar modules. Larger dust particles have a higher effect on lowering solar module efficiency.

The photovoltaic system installed in the case study building will annually provide the building with 160 819 kWh, which accounts for 10.5% of the annual required energy for the building. The annual reduction of photovoltaic power due to the accumulation of dust particles on cells is 89.33%, so the installed photovoltaic system will annually provide the building with 17 091.314 kWh, which accounts for 1.1% of the annual required energy of the building. After being exposed to the environmental conditions in its location, the IoT-based automatic solar panel cleaner will enhance the power generation of the installed system to generate 160 014.9 kWh, which means that it nearly returned to its ideal performance and generated 99.4% of the system's power original output.

It was discovered that the solar panel's efficiency rose over the summer season, which ran from June to September. Solar panel efficiency decreased significantly during the winter season, which ran from December to March. As measurements were taken biweekly, allowing dust to accumulate on the cells even during months with low dust levels and better conditions, it became noticeable that by the end of each 2-week period, a significant amount of dust had built up on the panels, which reduced the variation across different moths.

The accumulation of dust on the surface of a photovoltaic cell has a negative impact on its efficiency. As the amount of dust increases, the energy generated by the cell decreases. Hence, the efficiency reduction (η reduction) of photovoltaic panels when they are covered with dust can be calculated using equation (5),

$$\eta_{\text{reduction}} = \frac{\eta_{\text{normal}} - \eta_{\text{dirty}}}{\eta_{\text{normal}}} 100\% \tag{5}$$

- η_{normal} represents the efficiency of the panel's elementary state.
- η_{dirty} represents the efficiency of the panel covered with dust.

The efficiency reduction caused by the accumulation of dust may be calculated by subtracting the dirty panel's efficiency from the normal panel's efficiency. To ensure high efficiency in the conversion of solar energy, photovoltaic modules must be cleaned on a regular basis, taking into consideration air pollution. To compute the efficiency recovered (η recovered) of solar panels after dust cleaning, use equation (6).

$$\eta_{\text{reduction}} = \frac{\eta_{\text{normal}} - \eta_{\text{clean}}}{\eta_{\text{normal}}} 100\% \tag{6}$$

It is vital to note that these equations give a general framework for estimating solar panel efficiency as well as dust-

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Table 5. Readings of solar cell before cleaning, after installation (2 weeks), and after cleaning

Readings	Date	Normal (clean)			Dirty solar cell			Cleaned solar cell		
		Voltage (V)	Current (A)	Power (W)	Voltage (V)	Current (A)	Power (W)	Voltage (V)	Current (A)	Power (W)
1	1 January 2023	22.50	0.55	12.35	19.83	0.33	6.49	22.48	0.55	12.29
2	15 January 2023	22.54	0.54	12.37	19.87	0.33	6.50	22.52	0.54	12.32
3	29 January 2023	23.01	0.56	12.63	20.28	0.33	6.64	22.99	0.56	12.58
4	12 February 2023	23.86	0.58	13.10	21.03	0.35	6.88	23.84	0.58	13.07
5	26 February 2023	23.90	0.58	13.12	21.06	0.35	6.89	23.88	0.58	13.05
6	11 March 2023	24.46	0.60	13.43	21.56	0.36	7.06	24.44	0.59	13.36
7	25 March 2023	25.23	0.61	13.85	22.25	0.39	7.28	25.21	0.61	13.78
8	8 April 2023	27.95	0.68	15.34	24.63	0.41	8.06	27.93	0.67	15.27
9	22 April 2023	31.01	0.76	17.02	27.33	0.45	8.94	30.98	0.75	16.94
10	6 May 2023	33.29	0.81	18.27	29.34	0.48	9.60	33.26	0.81	18.18
11	20 May 2023	35.70	0.87	19.60	31.46	0.52	10.30	35.67	0.87	19.53
12	3 June 2023	36.90	0.90	20.25	32.52	0.54	10.64	36.87	0.90	20.16
13	17 June 2023	40.04	0.98	21.98	35.29	0.58	11.55	40.00	0.97	21.87
14	1 July 2023	43.26	1.05	23.74	38.13	0.63	12.48	43.22	1.05	23.63
15	15 July 2023	48.02	1.17	26.36	42.32	0.70	13.85	47.98	1.16	26.23
16	29 July 2023	25.19	0.61	13.83	22.20	0.37	7.27	25.17	0.61	13.76
17	12 August 2023	46.88	1.14	25.73	41.32	0.68	13.52	46.84	1.14	25.61
18	26 August 2023	45.47	1.11	24.96	40.07	0.66	13.12	45.43	1.10	24.84
19	9 September 2023	43.12	1.05	23.67	38.00	0.63	12.44	43.08	1.05	23.51
20	23 September 2023	39.90	0.97	21.90	35.17	0.58	11.51	39.86	0.97	21.79
21	7 October 2023	38.09	0.93	20.91	33.57	0.55	10.99	38.05	0.92	20.81
22	21 October 2023	35.04	0.85	19.23	30.88	0.51	10.11	35.00	0.85	19.15
23	4 November 2023	32.93	0.80	18.07	29.02	0.48	9.50	32.90	0.80	17.99
24	18 November 2023	29.45	0.72	16.16	25.96	0.43	8.49	29.42	0.71	16.05
25	2 December 2023	25.80	0.63	14.16	22.74	0.37	7.44	25.79	0.63	14.09
26	16 December 2023	24.28	0.59	13.33	21.40	0.35	7.00	24.26	0.59	13.28
27	30 December 2023	22.42	0.55	12.31	19.76	0.33	6.47	22.41	0.54	12.26

induced efficiency decline before and after cleaning. Table 5 shows the readings of the solar cell before cleaning, 2 weeks after installation, and after cleaning. It is also noticed that after cleaning, the system reached an average of 99.5% of the cell's original output as the system does not remove all the accumulated dust.

Next, we investigated how dust deposition on solar module surfaces affects their performance. Fig. 10 depicts the energy decrease of solar panels during a 1-year period (January 2023–December 2023) as well as the solar panel efficiency drop before and after the cleaning process, as calculated by equations (5) and (6).

According to the measurements and readings, the photovoltaic cells' power dropped to almost half after the dust accumulated on it for 2 weeks, yet the cleaning system raised its power output again to almost 99.5% of its original output. The power output of the cells and electricity production are reduced because dust particles on the surface of the cells block and scatter sunlight, reducing the amount of sunlight reaching the photovoltaic material. However, under the same dust conditions, the voltage of the photovoltaic cells dropped by only 88% due to the fundamental properties of the solar cell materials and construction that is not affected by reduced light absorption. The power output of the cells measured in watts is the most important factor in generating electricity from photovoltaic as it represents the actual rate of energy production and the amount of work that can be performed by the electricity being generated.

6 Conclusions and Recommendations

Photovoltaic solar cell systems have recently been developed to generate electricity from the sun and substitute fossil fuel in generating electricity. It has significant potential especially in the MENA region due to its great solar potential. Using solar power instead of fossil fuel to generate electricity reduces carbon dioxide emissions greatly. Regions like the MENA region are characterized by the desert climate, which features dust pollution. Accumulation of dust on the photovoltaic panels has a significant impact on their output and performance. The experimental study was carried out in a period of 1 year (January 2023–December 2023) in Cairo, Egypt, to test the efficiency of the proposed IoT cleaning system of photovoltaics. According to the results, the accumulation of dust on the cells resulted in an 89.33% reduction in the cell power, a 40.3% reduction in the cell current, and 11.8% in its voltage. The cell with the proposed IoT cleaning system installed recovered the power output of the cell to 99.4% of its original power output.

According to the obtained results from the applied study, installing the IoT-based automatic solar panel cleaner in an office building located in Cairo, Egypt, resulted in keeping the solar panels' power output at 99.4% of its original power output, providing the building with 10.5% of its annual energy consumption.

It is recommended to apply further research on other systems that aim to enhance the efficiency of solar panels, such as drone aerodynamic systems and other IoT-based automated solar panel cleaning. Drones, equipped with advanced sensors and cleaning mechanisms, offer a promising solution for the maintenance and cleaning of solar panels, particularly in large-scale installations and hard-to-reach areas. These systems can provide continuous, automated cleaning, reducing the need for manual intervention and ensuring that panels remain at their optimal performance levels. In addition to drone-based solutions, IoT-based

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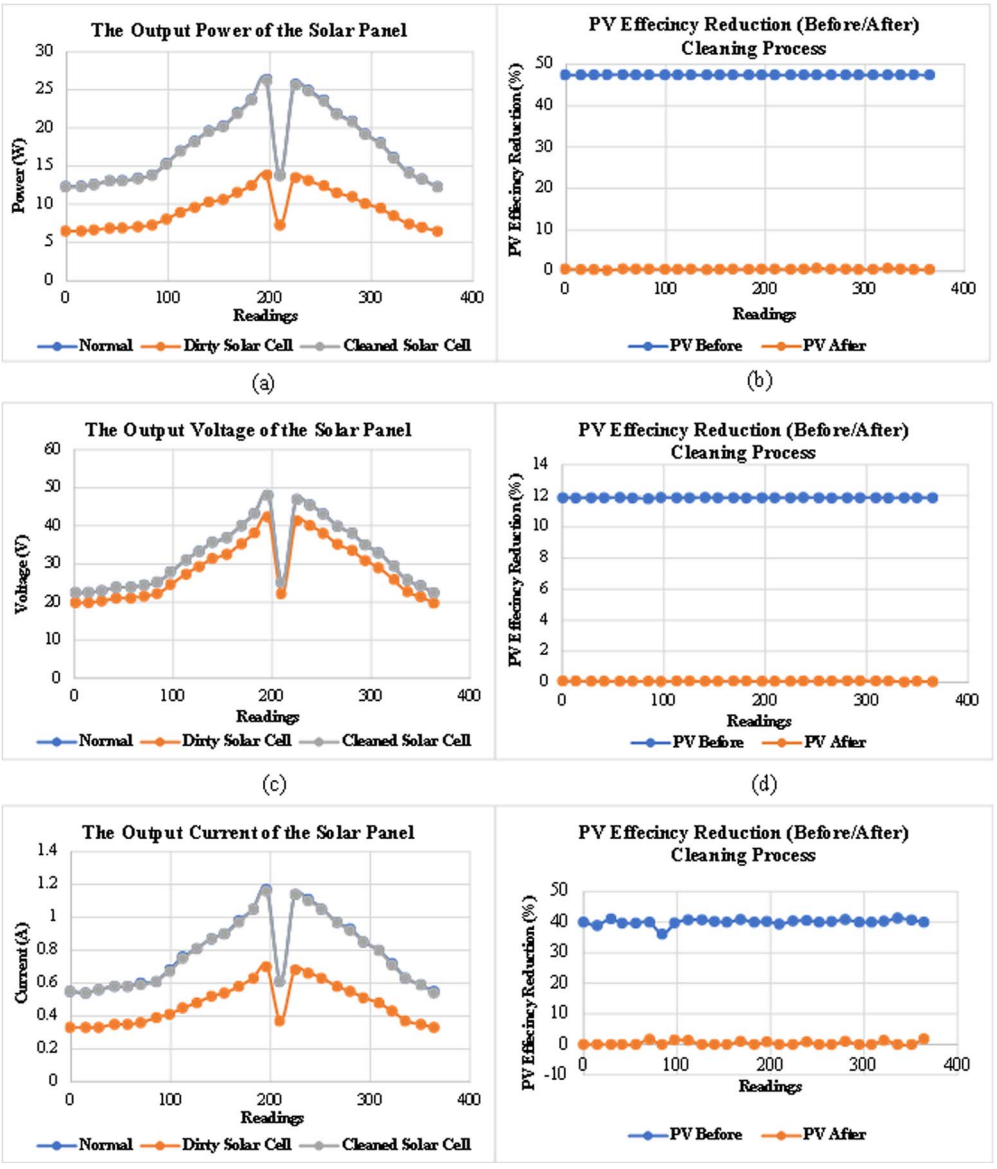


Figure 10. Solar panel behavior over 1 year. (a) Power losses of solar panels in 2 weeks. (b) Decrease in photovoltaic panel efficiency owing to dust cover. (c) Voltage losses of solar panels in 2 weeks. (d) Reduction ratio in solar panel voltage due to dust accumulation. (e) Current losses of solar panels in 2 weeks. (f) Reduction ratio in solar panel current due to dust accumulation.

automated cleaning systems that utilize real-time data and machine learning algorithms should be explored. These systems can dynamically schedule cleaning activities based on environmental conditions and panel performance data, thereby maximizing efficiency and reducing operational costs. Furthermore, it is also recommended to conduct extensive research on control and monitoring systems and their impact on the efficiency of solar panels across different climatic conditions. Effective control systems are essential for optimizing the performance of solar installations, and understanding their effectiveness in various environments is crucial. This research is particularly important for regions characterized by large desert areas where dust accumulation is a significant issue. In such areas, innovative dust mitigation strategies, such as electrostatic cleaning and the application of hydrophobic coatings, should be investigated. These strategies can help maintain the cleanliness of solar panels, thus ensuring sustained efficiency and performance. Additionally, developing robust algorithms for real-time monitoring and predictive maintenance is necessary. These algorithms should be capable of analyzing performance data to predict potential issues and schedule maintenance activities before significant efficiency losses occur. Such predictive maintenance systems can help reduce downtime and maintenance costs, thereby enhancing the overall reliability and profitability of solar installations. Research should also consider the integration of these advanced systems into existing solar infrastructures, ensuring that they can be implemented seamlessly and cost-effectively. Moreover, it is important to evaluate the environmental and economic impacts of these advanced systems. Conducting life cycle assessments and cost–benefit analyses will provide valuable insights into the long-term sustainability and financial viability of drone-based and IoT-based cleaning systems, as well as advanced control and monitoring solutions. This holistic approach will help stakeholders make informed

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decisions about adopting these technologies in their solar projects. In conclusion, advancing research in drone aerodynamic systems, IoT-based automated cleaning, and sophisticated control and monitoring systems is essential for enhancing the efficiency of solar panels. This research should focus on addressing the unique challenges posed by different climatic conditions, particularly in dust-prone desert regions. By developing and implementing these advanced technologies, we can ensure that solar energy systems operate at their maximum potential, contributing to the broader goal of sustainable and reliable renewable energy production.

Author contributions

Maha Anber (Conceptualization [equal], Data curation [equal], Formal analysis [equal], Investigation [equal], Methodology [equal], Project administration [equal], Resources [equal], Software [equal], Supervision [equal], Validation [equal], Visualization [equal], Writing—original draft [equal], Writing – review & editing [equal]) and Bassam Aboshosha (Conceptualization [equal], Data curation [equal], Formal analysis [equal], Investigation [equal], Methodology [equal], Project administration [equal], Resources [equal], Software [equal], Supervision [equal], Validation [equal], Visualization [equal], Writing—original draft [equal], Writing—review & editing [equal]).

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REFERENCES

1. Anber M. The efficiency of using mineral insulating solutions in buildings in Egypt. *Int Des J* 2024;**14**:371–9. https://doi.org/10.21608/idj.2023.249201.1099.
2. David N, Abioye O. Solar power system: a viable renewable energy source for Nigeria. *JECER* 2013;**1**:10–9.
3. Anber M. The efficiency of using building integrated photovoltaics in the retrofitting of office buildings in Egypt. *Port-Said Eng Res J* 2021;**0**:53–9. https://doi.org/10.21608/pserj.2021.64601.1095.
4. Mazumder MK, Sharma R, Biris AS. *et al.* Electrostatic Removal of Particles and its Applications to Self-Cleaning Solar Panels and Solar Concentrators. In R. Kohli, K.L. Mittal (eds) *Developments in Surface Contamination and Cleaning*. William Andrew, Elsevier Inc, 2011, 149–99.
5. Kutiel H. Dust storms in the Middle East: sources of origin and their temporal characteristics. *Indoor Built Environ* 2003;**12**:419–26. https://doi.org/10.1177/1420326X03037110.
6. Gholami A, Saboonchi A, Alemrajabi AA. Experimental study of factors affecting dust accumulation and their effects on the transmission coefficient of glass for solar applications. *Renew Energy* 2017;**112**:466–73. https://doi.org/10.1016/j.renene.2017.05.050.
7. Mani M, Pillai R. Impact of dust on solar photovoltaic (PV) performance: research status, challenges and recommendations. *Renew Sustain Energy Rev* 2010;**14**:3124–31. https://doi.org/10.1016/j.rser.2010.07.065.
8. Salim A, Huraib F, Eugenio N. PV power—study of system options and optimization. In *Proceedings of the 8th European PV Solar Energy Conference*. Engineering, Environmental Science, 1988.
9. Wakim F. *Introduction of PV power generation to Kuwait*. Kuwait: Kuwait Institute for Scientific Researchers, 1981.
10. Sayigh AAM, Al-Jandal S, Ahmed H. Dust effect on solar flat surfaces devices in Kuwait. *Workshop on the Physics of Non-conventional Energy Sources and Materials Science for Energy*. Triest, Italy: the International Centre for Theoretical Physics (ICTP), 1985.
11. Kazem HA, Khatib T, Sopian K. *et al.* Effect of dust deposition on the performance of multi-crystalline photovoltaic modules based on experimental measurements. *Int J Renew Energy Res* 2013;**3**:850–3.
12. Rahman M, Islam A, Karim AH. *et al.* Effects of natural dust on the performance of PV panels in Bangladesh. *Int J Mod Educ Comput Sci* 2012;**10**:26–32. https://doi.org/10.5815/ijmecs.04.10.2012.
13. Shaharin AS, Haizatul HH, Nik Siti HN. *et al.* Effects of dust on the performance of PV panels. *Int J Mech Ind Mechatron Manuf Eng* 2011;**5**:588–93.
14. Ali HM, Zafar MA, Bashir MA. *et al.* Effect of dust deposition on the performance of photovoltaic modules in Taxila, Pakistan. *Therm Sci.* 2017;**21**:915–23. https://doi.org/10.2298/TSCI140515046A.
15. Tanesab J, Parlerliet D, Whale J. *et al.* The contribution of dust to performance degradation of PV modules in a temperate climate zone. *Sol Energy* 2015;**120**:147–57. https://doi.org/10.1016/j.solener.2015.06.052.
16. Kaldellis JK, Kokala A. Quantifying the decrease of the photovoltaic panels' energy yield due to phenomena of natural air pollution disposal. *Energy* 2010;**35**:4862–9. https://doi.org/10.1016/j.energy.2010.09.002.
17. Hee JY, Kumar LV, Danner AJ. *et al.* The effect of dust on transmission and self-cleaning property of solar panels. *Energy Procedia* 2011;**15**:129–34. https://doi.org/10.1016/j.egypro.2012.02.015.
18. Farah A, Esra K, Mohamed O. Effect of dust on the efficiency of solar PV panel in Khartoum. *Proceedings of the 6th International Conference on Energy Harvesting, Storage, and Transfer (EHST)*. Orléans, Ontario, Canada: International ASET Inc., 2022.
19. Walaman-i IS, Beem H. Design of a dust cleaning machine to reduce dust soiling on solar PV panels in Ghana. *IEEE Global Humanitarian Technology Conference (GHTC)* 2022;481–4.
20. Inayah I, Hayati N, Nurcholis A. *et al.* Realtime monitoring system of solar panel performance based on internet of things using Blynk application. *ELINVO* 2022;**7**:135–43. https://doi.org/10.21831/elinvo.v7i2.53365.
21. Kumar D, Singh B, Chandra A. Wireless sensor network-based system for monitoring and fault detection of solar panels. *Proceedings of the International Conference on Inventive Communication and Computational Technologies* 2017;746–52.
22. Zhou L, Jiang Y, Zhang X. *et al.* An AI-based solar panel cleaning system. *J Renew Energy* 2021;**177**:275–81.
23. Tewari A, Sharma A, Bhattacharya S, Singhal M. IoT-based solar panel monitoring and cleaning system using machine learning algorithms. *Proceedings of the International Conference on Computing, Power and Communication Technologies*. Piscataway, NJ, USA: IEEE, 2020, 1–6.
24. Bhat MA, Choudhary MA, Ahmad F. *et al.* Development of drone-based solar panel cleaning and monitoring system. *J Renew Energy* 2021;**173**:855–64.
25. Kippelen B, Faria LJ, de Oliveira LFC. *et al.* Hydrophobic coating for solar panels: a study of water repellency and efficiency. *J Renew Sustain Energy Rev* 2020;**130**:109926.
26. Aljaghoub H, Abumadi F, AlMallahi MN. *et al.* Solar PV cleaning techniques contribute to sustainable development goals (SDGs) using multi-criteria decision-making (MCDM): assessment and review. *Int J Thermofluids* 2022;**16**:100233. https://doi.org/10.1016/j.ijft.2022.100233.
27. Omran M. Analysis of solar radiation over Egypt. *Theor Appl Climatol* 2000;**67**:225–40. https://doi.org/10.1007/s007040070011.
28. *Global Solar Atlas*. www.globalsolaratlas.info.
29. Robaa S. A study of solar radiation climate at Cairo urban area, Egypt and its environs. *Int J Climatol* 2006;**26**:1913–28. https://doi.org/10.1002/joc.1349.
30. Mayhoub MS, Elqattan AA, Algendy AS. Experimental investigation of dust accumulation effect on the performance of tubular daylight guidance systems. *Renew Energy* 2021;**169**:726–37. https://doi.org/10.1016/j.renene.2021.01.061.

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